BONUS REPORT · COMPANION TO AFRICA'S AI MOMENT

AFRICA–G7 AI PREPAREDNESS GAP

2026 Digital Divide Audit — 10 African economies measured against all seven G7 nations, and the five investments that close the distance.

0.34
SUB-SAHARAN AFRICA

0.72
G7 AVERAGE

47%
AFRICA'S INFRASTRUCTURE VS G7

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AFRICAN SAMPLE
South Africa, Nigeria, Kenya, Egypt, Ghana, Morocco, Algeria, Ethiopia, Tanzania, Zambia

PRIMARY SOURCES
IMF AI Preparedness Index (AIPI) · IMF 2026 Africa research · Stanford HAI 2026 AI Index Report

EXECUTIVE SUMMARY

THE AI GAP IS ABOUT TO BECOME AN INCOME GAP.

The AI divide between Africa and the G7 is already large enough to become a productivity, income, and industrial-ownership divide if it is not addressed quickly.

0.72	0.34	0.39	0.62
G7 GROUP SCORE	SUB-SAHARAN AFRICA	10-COUNTRY SAMPLE AVG.	ITALY — LOWEST G7

The IMF's latest available AI Preparedness Index is based on 2023 data. It scores the G7 as a group at 0.72, compared with 0.34 for Sub-Saharan Africa. Across the 10 African economies examined here, the average is approximately 0.39.

FIVE FINDINGS STAND OUT

01 **Even Africa's strongest countries remain well behind the G7.** South Africa leads this sample at 0.50, followed by Kenya at 0.45 and Ghana and Morocco at 0.43. Italy, the lowest-scoring G7 member, is still at 0.62.

02 **Digital infrastructure is one of the two largest gaps.** The selected African economies average roughly 34/100 on a normalized version of the IMF infrastructure component versus about 72/100 across the G7.

03 **Regulatory and institutional capacity is equally problematic.** The African sample averages roughly 39/100 versus 78/100 for the G7. The issue is not simply whether countries have AI strategies; it is their ability to implement data governance, competition rules, cybersecurity, procurement standards and effective oversight.

04 **Human capital is Africa's strongest relative pillar, but it is insufficient by itself.** The selected economies average approximately 43/100 versus 66/100 for the G7. Stanford notably identifies South Africa as one of the few countries where engineering-oriented AI skills have recently been growing faster than general AI literacy.

05 **The global AI industry is becoming extremely concentrated.** Stanford reports that the United States hosts 5,427 data centers (more than ten times any other country) while U.S. private AI investment reached $285.9 billion in 2025. Industry produced more than 90% of notable frontier models in 2025, while frontier-model production remains heavily concentrated in the United States and China.

> *Africa does not need to replicate Silicon Valley or train the world's largest models. It does need enough electricity, broadband, compute, skills, capital, data and institutional capacity to adopt and adapt AI at scale.*

The IMF estimates that, at current preparedness levels, AI might add only **0.2%** to Sub-Saharan African GDP over the coming decade. With stronger foundations and broader adoption, the gain could rise to approximately **4%** over the decade.

1 · METHODOLOGY AND IMPORTANT CAVEAT

HOW THE NUMBERS WERE BUILT.

The IMF AIPI measures four dimensions: digital infrastructure; human capital and labor-market policies; innovation and economic integration; and regulation and ethics. It covers 174 economies and is designed as a measure of AI *adoption* preparedness rather than frontier-model invention capacity.

The IMF component values contribute up to approximately 0.25 each to the 0–1 overall AIPI. To make comparison easier, this audit converts the components to a 0–100 normalized scale, where 100 represents the maximum possible pillar contribution. This rescaling changes presentation, not relative performance.

Stanford's AI Index is different. It does not provide the same four readiness scores for every country. It is therefore used here as an ecosystem overlay — showing current trends in compute, investment, talent, education, adoption and regulation — rather than as a second country-ranking system.

Important: the IMF country scores remain a 2023 snapshot. The Stanford 2026 Index and IMF's July 2026 Africa research are used to update the interpretation of that baseline.

2 · AFRICAN AI PREPAREDNESS SCORECARD

IMF AIPI, normalized to 0–100 for comparison

AFRICAN ECONOMY	OVERALL AIPI	DIGITAL INFRA.	HUMAN CAPITAL	REGULATION & ETHICS	INNOVATION & INTEG.	MAIN BOTTLENECK
South Africa	50	48	48	56	44	Innovation, scale & diffusion
Kenya	45	40	44	52	45*	Infrastructure & advanced skills
Morocco	43	40	48	40	43*	Regulation & compute
Ghana	43	35	45	54	45*	Infrastructure & commercialization
Egypt	39	36	48	32	39*	Regulatory environment
Algeria	37	32	56	28	39*	Innovation & regulation
Zambia	37	28	44	36	36*	Infrastructure
Tanzania	35	24	36	36	32*	Infrastructure & skills
Nigeria	34	32	36	28	32*	Infrastructure & regulation
Ethiopia	25	20	24	28	24*	Broad foundational capacity
Sample average	**39**	**34**	**43**	**39**	**38**	—

Overall AIPI country values are from the IMF DataMapper. The IMF's published annex independently shows the component structure for several of these economies, including Egypt, Morocco, Nigeria and South Africa. Component figures for several additional African economies are reproduced from the IMF dashboard in the published Systems dataset; Ghana's individual component series are also available through data mirrors of the IMF/World Bank series. *Innovation & integration values marked with an asterisk are the average of that country's other three pillar scores, pending the individually reported figure.

WHAT THE COUNTRY RESULTS REVEAL

TEN ECONOMIES, ONE PATTERN.

South Africa is the continental benchmark in this sample, but a 0.50 AIPI is still well below every G7 country. Its relatively strong regulatory score and recent growth in AI engineering skills suggest it can become a regional AI-production and compute hub, but innovation financing, firm scaling and diffusion beyond large corporations remain critical.

Kenya has one of the most balanced African profiles. Its relatively high regulatory score provides an institutional base, but digital infrastructure and deeper technical capacity still constrain its ability to turn its technology ecosystem into economy-wide AI productivity.

Ghana's human-capital and regulatory components are comparatively strong, while infrastructure and innovation are weaker. This is a classic case where policy readiness could outrun the physical and commercial capacity required to execute it.

Egypt and Algeria possess significant human-capital potential but show large regulatory and innovation gaps. Their challenge is less simply producing educated workers and more turning those workers into competitive AI firms, research activity and productive deployment.

Nigeria illustrates the scale paradox. It possesses one of Africa's largest markets and technology ecosystems, yet its AIPI of roughly 0.34 is only around the Sub-Saharan African regional average. Electricity, connectivity, institutional capacity and broad digital adoption therefore matter as much as startup activity.

Ethiopia, Tanzania and Zambia demonstrate why basic digital foundations cannot be bypassed. Where infrastructure scores remain very low, sophisticated national AI strategies by themselves will have limited economic effect.

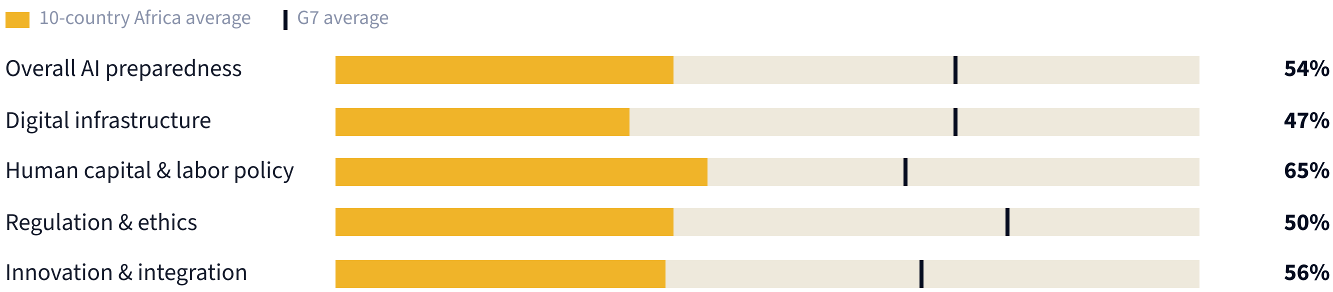
3 · THE G7 BENCHMARK

G7 ECONOMY	OVERALL AIPI	DIGITAL INFRA.	HUMAN CAPITAL	REGULATION & ETHICS	INNOVATION & INTEG.
United States	77	76	72	88	72
Germany	75	76	72	80	72
United Kingdom	73	72	68	84	64
Japan	73	72	68	80	72
Canada	71	68	68	84	64
France	70	72	64	72	68
Italy	62	68	52	60	64
G7 average	**72**	**72**	**66**	**78**	**68**

The IMF itself reports the G7 aggregate AIPI at approximately 0.72. Individual country and component values are also reported in the IMF's country annex.

The striking point is not simply that the G7 performs better. It is the **consistency of the enabling environment**. Italy, the lowest G7 performer overall, has a digital-infrastructure score equivalent to roughly 68/100 — substantially higher than South Africa's 48 and more than double Nigeria's 32.

AFRICA RUNS AT HALF OF G7 CAPACITY.



DIMENSION	AFRICA AVG.	G7 AVG.	AFRICA AS % OF G7	APPROX. GAP
Overall AI preparedness	39	72	54%	33 pts
Digital infrastructure	34	72	47%	38 pts
Human capital & labor policy	43	66	65%	23 pts
Regulation & ethics	39	78	50%	39 pts
Innovation & integration	38	68	56%	30 pts

Two structural weaknesses are especially severe: **digital infrastructure**, where African capacity in the sample is less than half the G7 benchmark, and **regulatory and institutional capacity**, at approximately half the G7 level. Human capital performs better in relative terms, but a skills pipeline without sufficient compute, reliable infrastructure, firms, finance and effective institutions can produce brain drain rather than domestic productivity growth.

COMPUTE, CAPITAL AND TALENT ARE CONCENTRATING.

A. Compute is becoming concentrated

Stanford estimates global AI compute capacity reached 17.1 million H100-equivalent GPUs, expanding roughly 3.3× per year since 2022. The United States alone hosts 5,427 data centers, more than ten times any other country. African countries that lack competitive compute will increasingly purchase AI capability from foreign cloud and model providers — successfully *using* AI while capturing only a small portion of the value it creates.

B. Capital concentration reinforces industrial concentration

U.S. private AI investment reached $285.9 billion in 2025, more than 23 times China's reported $12.4 billion. The U.S. also recorded 1,953 newly funded AI companies. Industry produced more than 90% of notable frontier models. Countries entering the cycle late face an increasingly difficult catch-up problem; Africa cannot rely solely on ordinary startup policy.

C. Human capital offers an opportunity

South Africa is one of only three countries highlighted by Stanford — alongside the UAE and Chile — where engineering-oriented AI skills have recently grown faster than general AI literacy. The lesson is to expand beyond "AI awareness": Africa needs more AI and software engineers, data engineers, cybersecurity and cloud specialists, researchers, AI product managers, technically capable regulators, and teachers who can deliver AI education. The objective is productive capacity, not merely prompt-engineering literacy.

D. Regulation is expanding; infrastructure is not keeping pace

More than half of newly adopted national AI strategies in 2024 came from emerging economies. Yet between 2018 and 2025, Europe and Central Asia expanded state-backed AI supercomputing clusters from 3 to 44, while the Middle East and North Africa reached only 8. Sub-Saharan Africa has simultaneously adopted 71 data-localization measures. The danger: increasingly restrictive national data regimes without the domestic compute to make them productive. The better approach is regional digital sovereignty, not 54 isolated sovereign technology stacks.

WHERE THE MONEY HAS TO GO.

1 Power, broadband, cloud and compute infrastructure CRITICAL

This is the foundational bottleneck. The IMF explicitly identifies reliable electricity, affordable broadband, data infrastructure and worker skills as prerequisites for widespread African AI adoption. Trying to build an AI economy without solving power and connectivity is equivalent to attempting industrialization without roads or electricity.

- Reliable power for tech & industrial zones
- Fiber backbones & open-access networks
- Rural & last-mile connectivity
- Internet exchange infrastructure
- Regional cloud & data-center capacity
- Shared GPU/HPC for universities & startups
- Affordable compute-credit programs for SMEs
- Data platforms for agriculture, health, transport, education, government

2 Human capital at mass scale, not just elite AI talent CRITICAL

Africa's human-capital score is the strongest of the four relative pillars, but still only around 65% of the G7 benchmark. Investment must operate at several levels simultaneously: **schools** (mathematics, computational thinking, data and AI literacy); **TVET and universities** (software engineering, data science, cloud, AI engineering, robotics, cybersecurity); **the existing workforce** (AI-enabled productivity rather than wholesale replacement); **government** (AI-literate procurement officers, regulators, cybersecurity staff, senior administrators); and **business owners and managers** (workflow redesign, automation, adoption). The goal is to move millions of workers from AI users to AI-augmented workers, while creating a smaller but strategically important population of AI creators.

3 Regulatory capacity and trusted institutions CRITICAL

Regulation represents the largest measured gap alongside infrastructure. Countries need more than national AI strategies or ethics statements. The IMF warns that AI can increase inequality when gains are concentrated among large companies, skilled workers and urban centers, and calls for practical rules covering data, competition, cybersecurity, consumer protection and public-sector AI.

- Data-protection authorities
- Cybersecurity & incident response
- Algorithmic auditing
- AI procurement standards
- Competition policy
- Consumer protection
- Digital identity governance
- Cross-border data frameworks
- Regulatory sandboxes
- Public-sector AI governance
- Judicial & regulatory technical expertise

OWNERSHIP AND DIFFUSION.

4 Domestic innovation, commercialization and growth capital HIGH

Graduates are not enough if the IP, companies and skilled workers subsequently migrate elsewhere. The strategic objective is not for every African country to develop its own frontier LLM — it is for Africa to own more of the applications, data, infrastructure, IP and companies built on top of global AI technology.

- African AI venture & growth funds
- University–industry research
- Regional research compute
- Commercialization grants
- Startup procurement programs
- Local-language datasets & African-language models
- Open-source African AI
- Diaspora collaboration incentives

5 Diffusion to SMEs, farms and the informal economy HIGH

This may ultimately determine whether AI reduces or increases inequality. If adoption stays concentrated among banks, telcos, large mines, multinationals and urban tech firms, aggregate adoption can rise while inequality worsens. The IMF argues Africa's largest gains may come from extending AI to farms, schools, clinics, small businesses, tax administration and informal enterprises. Africa's winning AI applications may look very different from those developed for high-income economies.

- AI adoption vouchers for SMEs
- Low-cost cloud credits
- Agricultural advisory systems
- Microbusiness bookkeeping
- Local-language voice interfaces
- Healthcare decision support
- Teacher copilots
- Mobile-first, low-bandwidth apps

7 · COUNTRY-LEVEL INVESTMENT PRIORITIES

WHAT EACH COUNTRY SHOULD DO FIRST.

COUNTRY	IMMEDIATE PRIORITY	SECOND PRIORITY	STRATEGIC OPPORTUNITY
South Africa	Scale innovation / compute	Advanced AI talent	Continental AI infrastructure hub
Kenya	Power, compute and connectivity	Advanced engineering skills	East African AI / services hub
Nigeria	Electricity & digital infrastructure	Governance / regulatory capacity	Large-market AI commercialization
Ghana	Compute / connectivity	Innovation finance	West African responsible-AI hub
Egypt	Regulatory modernization	Commercialization	Arabic-language / industrial AI
Morocco	Regulation & compute	R&D scaling	Manufacturing and industrial AI
Algeria	Innovation ecosystem	Regulatory modernization	Convert human capital into technology firms
Tanzania	Connectivity & power	Workforce skills	Agriculture / logistics AI
Zambia	Digital infrastructure	Regulatory capacity	Mining / agriculture AI
Ethiopia	Basic digital foundations	Human capital	Agriculture / public-service AI

These priorities are diagnostic interpretations of the IMF pillar structure rather than new IMF country recommendations.

8 · HOW THE AI DIVIDE COULD BECOME A PERMANENT INCOME DIVIDE

TWO CYCLES, PULLING APART.

The danger is a self-reinforcing cycle.

ADVANCED-ECONOMY CYCLE

Better infrastructure → faster AI adoption → higher productivity → higher wages / profits → more investment → better models and companies → further productivity gains

LAGGING-ECONOMY CYCLE

Poor infrastructure → expensive AI adoption → lower firm productivity → weaker profits / tax revenue → less investment in infrastructure and skills → even slower adoption

Stanford's finding that generative-AI adoption correlates strongly with GDP per capita suggests this feedback loop may already be emerging. Without intervention, AI could widen inequality at three levels at once: **between countries** (G7 and frontier economies gain productivity faster), **within African countries** (urban, educated, formal-sector workers gain more than rural and informal workers), and **between firms** (large corporations access compute, proprietary data and specialist talent that SMEs cannot afford).

9 · AFRICA NEEDS A REGIONAL STRATEGY

Many African countries individually lack the market size to justify large compute facilities, specialized research infrastructure or deep pools of AI capital. The IMF specifically emphasizes regional cooperation around infrastructure, standards, regulation and markets: regional GPU and supercomputing facilities, cross-border digital infrastructure, compatible data-governance frameworks, interoperable digital identities, shared cybersecurity standards, African-language datasets, regional research networks, cross-border startup markets, coordinated procurement and regional AI investment vehicles. This is where the African Continental Free Trade Area can become more than a goods-trade project.

> *The question is not "Can Nigeria, Ghana, Kenya or Rwanda individually compete with the United States?" It is "How can African economies combine their markets, talent, data, capital and infrastructure so African companies can compete globally?"*

10 · RECOMMENDED INVESTMENT SEQUENCE, 2026–2035

THE FIVE-YEAR PLAN, AND THE FIVE AFTER IT.

2026–2029
Phase I

Build the foundation

Prioritize power reliability, broadband affordability, government digitization, cloud access, cybersecurity, basic AI literacy, public-sector data infrastructure and practical AI governance.

2028–2032
Phase II

Build productive capacity

Expand AI engineering education, shared regional compute, sector datasets, university R&D, commercialization capital and AI adoption among SMEs.

2030–2035
Phase III

Capture value

Scale African AI companies across borders, develop exportable AI products, expand African-language technologies, deepen advanced research and increase African ownership of compute, platforms and intellectual property. The three phases should overlap rather than occur sequentially.

11 · WHAT GOVERNMENTS SHOULD MEASURE EVERY YEAR

A useful African AI-readiness dashboard should go beyond counting national strategies or startups:

- Electricity reliability for firms and data infrastructure
- Broadband cost relative to household income
- Fiber and 4G/5G coverage
- Domestic and regional compute availability
- Cost of GPU/cloud access for startups
- % of schools teaching computing and AI
- AI/STEM graduates per 100,000 people
- Advanced AI-engineering workforce
- AI adoption among large firms vs SMEs
- Rural vs urban AI adoption
- Domestic AI investment
- Local AI patents, research, open-source contributions
- Public datasets available through APIs
- Government AI deployment
- Regulatory enforcement capability
- Cross-border data interoperability
- **% of AI spending captured by domestic or African companies**

The last metric is particularly important. AI adoption alone should not be treated as success. **Value capture matters.**

CONCLUSION

CONSUMER, OR CREATOR.

Africa's AI problem is not primarily a lack of enthusiasm for artificial intelligence. It is a shortage of the systems that convert AI into productive economic capacity.

The 10 African economies examined average only about 54% of G7 overall preparedness. Digital infrastructure is around 47% of the G7 benchmark, regulatory capacity about 50%, innovation capability about 56%, and human capital approximately 65%. At the same time, compute, capital, frontier models and major AI companies are becoming extremely concentrated internationally. That creates a narrow window.

Africa does not need to win the race to build the world's largest foundation models. It does need to ensure that its people and businesses are not relegated to being permanent renters of foreign intelligence, foreign compute and foreign digital infrastructure.

Power and connectivity → compute and data → skills → trusted institutions → innovation capital → mass adoption → African ownership and regional scale

If that sequence is pursued aggressively, AI can help Africa leapfrog constraints in agriculture, healthcare, education, finance, government and enterprise. If it is not, the AI revolution could deepen precisely the disparities it promises to solve. The IMF's latest assessment puts the stakes in measurable terms: about **0.2%** additional output over a decade under today's trajectory versus roughly **4%** if the foundations for broad AI adoption are successfully built.

The central challenge for African policy is no longer whether AI is coming. It is whether Africa will merely consume the intelligence produced elsewhere or build enough capability, ownership and scale to participate in the wealth it creates.

THE REPORT IS THE DIAGNOSIS. THE BOOK IS THE PLAN.

AFRICA'S AI MOMENT

From Consumers of Technology to Creators of the Future

Seventeen chapters on how Africans can build, lead, and prosper in the age of artificial intelligence — education, healthcare, agriculture, jobs, entrepreneurship, AI sovereignty, and what the diaspora can remit beyond money. Built around the BUILD framework: Build African capability, Use AI to solve African problems, Invest in African innovation, Lead responsibly, Develop African ownership.

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